



Subject Name-WEB DESIGNING

Course Code:-BECS25-103

UNIT-D

Adobe Photoshop and JAVA Script



UNIT- D TOPICS- 1

- **Stock Photography**
- **Types of image graphics**
- **Vector and Scalar graphics and tools**
- **Colour modes and resolution options**
- **Filter Effects**
- **Animation using Photoshop.**



STOCK PHOTOGRAPHY

Definition:

- Stock photography refers to professionally captured images that are available for licensing and use in marketing, design, websites, publications, etc.
- Photographers upload images to agencies, where users can license them for a fee, or sometimes find them for free through platforms like [Canva](#).
- This can be a source of income for photographers and a convenient way for designers to access visuals without hiring a photographer for each project.



STOCK PHOTOGRAPHY

Definition:

Stock photography refers to **pre-shot professional images** that are **licensed for specific uses** — such as advertisements, websites, blogs, magazines, and other media.

Instead of hiring a photographer for every project, companies or designers can buy **ready-made stock photos** that fit their needs.

These images are stored in large online **stock libraries or agencies**, such as:

Shutterstock

Adobe Stock

Getty Images

iStock

Pexels

Unsplash

Pixabay



STOCK PHOTOGRAPHY

How it works

- **For photographers:** Photographer take photos (or videos) and upload them to a stock agency's platform. They then earn a commission or royalty when someone licenses an image you created.
- **For users:** User can search the database of images and purchase a license to use a photo for your project, such as a blog post, advertisement, or brochure, instead of hiring a photographer.
- **Licensing:** "Royalty-free" stock photos can be used multiple times after acquiring a license, without paying an extra fee for each use.
- **Content:** Stock photos are available for a wide range of subjects, from professional models in specific scenarios to everyday objects and processes, notes.



STOCK PHOTOGRAPHY

Type	Description	Example Use
Royalty-Free (RF)	Pay once, use multiple times (no time limits). Not “free,” but flexible.	Website backgrounds, ads, brochures
Rights-Managed (RM)	License for specific use (duration, location, medium). Pricing depends on scope.	Exclusive ad campaigns
Editorial Use Only	Used for news, blogs, or educational content; cannot be used commercially.	News article on a celebrity
Creative/Commercial Use	Used in advertisements, designs, packaging, etc. Requires model and property releases.	Product packaging, billboards



Types of image graphics

Digital images are visual representations created and displayed on computers or other digital devices.

They fall into **two main categories** based on how they store and display visual data:

- **Raster (Bitmap) Graphics**
- **Vector Graphics**

- The two main types of **image graphics** are **raster (pixel-based)** and **vector (path-based)**.
- Common raster formats include **JPEG, PNG, and GIF**, while popular vector formats include **SVG and EPS**.



Types of image graphics

RASTER GRAPHICS

- Raster graphics (also known as **bitmap graphics**) are made up of a **grid of tiny squares called pixels**.
- Each pixel represents a **single color**, and when combined, they form a **complete image**.
- When you zoom in on a raster image, you can see the **individual pixels** — this is why these images **lose clarity when enlarged**.

Structure:

Raster images are **resolution-dependent**, meaning their quality depends on the number of pixels (width × height).

Example: An image of **1920 × 1080 pixels** has 2,073,600 pixels total.

Higher pixel counts = higher quality and larger file sizes.



Types of image graphics

RASTER GRAPHICS

- **Description:** Made of a grid of pixels, these graphics are resolution-dependent. They are ideal for detailed images like photographs.
- **Pros:** Excellent for complex images with many colors and details.
- **Cons:** Lose quality when scaled up, becoming blurry or "pixelated".
- **Common formats:**
 - **JPEG** (.jpg): Great for photos due to its high compression for a smaller file size.
 - **PNG** (.png): Supports transparency, making it ideal for web graphics and icons.
 - **GIF** (.gif): Supports simple animations and a limited color palette.
 - **TIFF** (.tif): Used for high-quality printing and commercial applications, often uncompressed.
 - **BMP** (.bmp): A simple, often uncompressed format.
 - **WebP** and **AVIF**: Newer formats offering efficient compression.



Types of image graphics

RASTER GRAPHICS

Format	Full Form	Description / Use
JPEG (JPG)	Joint Photographic Experts Group	Compressed format for web photos; small file size but loses some quality.
PNG	Portable Network Graphics	Lossless compression; supports transparency; ideal for web graphics.
GIF	Graphics Interchange Format	Supports up to 256 colors; can be animated; used for icons or memes.
BMP	Bitmap Image File	Uncompressed; large file size; used in Windows systems.
TIFF (TIF)	Tagged Image File Format	Very high quality; used in printing, scanning, and publishing.
RAW	—	Captures unprocessed data from a camera's sensor; used by professional photographers.



Types of image graphics

RASTER GRAPHICS

Advantages:

- ✓ **High detail and realism** — perfect for photos and complex images.
- ✓ **Easy editing** of colors, brightness, and textures.
- ✓ **Widespread compatibility** — works with most software and platforms.
- ✓ **Supports a wide range of effects and filters** in programs like Photoshop.

Disadvantages:

- ✗ **Loses quality when scaled up (pixelation).**
- ✗ **Large file sizes**, especially for uncompressed or high-resolution images.
- ✗ **Limited scalability** — can't easily change shapes or sizes without distortion.
- ✗ **Editing individual elements is harder** — changes affect pixels, not objects.



Types of image graphics

RASTER GRAPHICS

Common Uses:

- Digital photography and photo retouching
- Web graphics and social media posts
- Digital paintings and textures
- Printing magazines, posters, and flyers
- Game art and visual effects

Editing Software (Raster-based):

- **Adobe Photoshop**
- **Corel PHOTO-PAINT**
- **GIMP (GNU Image Manipulation Program)**
- **Paint.NET**
- **Affinity Photo**



Types of image graphics

VECTOR GRAPHICS

Definition:

Vector graphics are made up of **mathematical equations and geometric primitives** (points, lines, curves, and shapes).

Unlike raster images, they are **not based on pixels** — they use **paths** that define the image.

This means they can be **scaled infinitely without any loss of quality**.

Structure:

Defined by **mathematical formulas** describing shapes and colors.

Each element (line, curve, shape) is a separate object that can be resized, moved, or recolored independently.

File size remains **small**, no matter the image dimensions.



Types of image graphics

VECTOR GRAPHICS

Format	Full Form	Description / Use
AI	Adobe Illustrator	Native file format for Illustrator; used in professional design.
SVG	Scalable Vector Graphics	Open standard for web-based vector images; supported by browsers.
EPS	Encapsulated PostScript	Used in printing and vector transfer between programs.
PDF	Portable Document Format	Can contain both vector and raster elements; great for print and sharing.
CDR	CorelDRAW	Native format for CorelDRAW software.



Types of image graphics

VECTOR GRAPHICS

Characteristics:

Resolution-independent — quality remains perfect at any size.

Created using **paths**, not pixels.

Easily editable — shapes, lines, and colors can be changed without affecting the rest of the image.

Best for **illustrations, logos, and typography**.



Types of image graphics

VECTOR GRAPHICS

Advantages:

- ✓ **Infinite scalability** — no loss of quality.
- ✓ **Smaller file sizes** than raster images.
- ✓ **Easily editable and customizable.**
- ✓ **Ideal for printing and large format designs.**
- ✓ **Perfect for clean, sharp designs (logos, icons, text).**

Disadvantages:

- ✗ Not suitable for **detailed or realistic** images like photographs.
- ✗ Requires **specialized software** to edit.
- ✗ Some older web browsers or apps may not support all vector formats.
- ✗ Complex designs can be harder to create compared to photo editing.



Types of image graphics

VECTOR GRAPHICS

Common Uses:

- Logo design and branding
- Icons, infographics, and diagrams
- Typography and text-based art
- Blueprints and CAD drawings
- Signage, banners, and billboards

Editing Software (Vector-based):

- **Adobe Illustrator**
- **CorelDRAW**
- **Inkscape** (free and open-source)
- **Affinity Designer**
- **Sketch** (for UI/UX design)



Types of image graphics

COMPARISON

Feature	Raster Graphics	Vector Graphics
Image Structure	Made of pixels	Made of paths (mathematical formulas)
Resolution	Resolution-dependent	Resolution-independent
Scalability	Loses quality when scaled	Infinite scalability
File Size	Larger	Smaller
Editing	Pixel-based	Shape/object-based
Best Used For	Photos, detailed images	Logos, icons, illustrations
Printing	Quality depends on DPI	Always sharp at any size
Software	Photoshop, GIMP	Illustrator, CorelDRAW
Web Support	JPEG, PNG, GIF	SVG
Complexity	Good for realism	Good for simplicity



Types of image graphics

CHOOSE THE RIGHT TYPE

Purpose	Recommended Type	Example File Format
Photography / Digital Art	Raster	JPEG, PNG, TIFF
Logo / Icon / Branding	Vector	AI, SVG, EPS
Website Graphics	Raster + Vector	PNG + SVG
Large-Scale Printing	Vector	EPS, PDF
Illustration / Infographics	Vector	SVG, AI
Magazine / Print Media	Raster	TIFF, JPEG



COLOUR MODES AND RESOLUTION OPTIONS

What Are Colour Modes?

Definition:

- A **colour mode** defines **how colors are represented and combined** in a digital image.
- It determines the **range of colors** that can be displayed or printed by combining primary colors in different ways.
- **Color** **modes** are the settings designers use to show colors consistently across devices and materials. Commonly used modes are **RGB, CMYK, index, grayscale, bitmap and LAB** which differ in quality and file size.
- Designers pick modes to optimize images and ensure these appear identically across media for brand consistency.



COLOUR MODES AND RESOLUTION OPTIONS

COLOR MODES

- **RGB** (Red, Green, Blue):

- Used for digital screens like monitors, websites, and apps.
- Uses additive color mixing to create a wide range of vibrant colors.

- **CMYK** (Cyan, Magenta, Yellow, Key/Black):

- Used for printed materials like brochures, magazines, and packaging.
- Uses subtractive color mixing, which works with inks.

- **Grayscale**:

- Represents images using shades of gray, from black to white.

- **Bitmap**:

- Displays images with only two colors: black and white.

- **Lab**:

- A device-independent model based on how humans perceive color.
- Useful for color correction and ensuring consistency across different media.

- **Indexed Color**:

- Uses a palette of up to 256 colors to reduce file size, making it suitable for web and multimedia.



COLOUR MODES AND RESOLUTION OPTIONS

COLOR MODES

• RGB (Red, Green, Blue):

- Used for digital screens like monitors, websites, and apps.
- Uses color mixing to create a wide range of vibrant colors.

RGB exists exclusively in digital formats (e.g., mobile screens). Although RGB is present across most electronic devices, the color elements vary across systems and models. So, an image can display differently on-screen from brand to brand.

Uses:

- Web & App Design.
- Digital Design.
- Social Media.
- Online Advertisements.





COLOUR MODES AND RESOLUTION OPTIONS

COLOR MODES

CMYK (Cyan, Magenta, Yellow, Key/Black): This *subtractive* color mode comprises Cyan, Magenta, Yellow and Key (Black), which combine to produce a range of hues. This four-color process works for most printers.

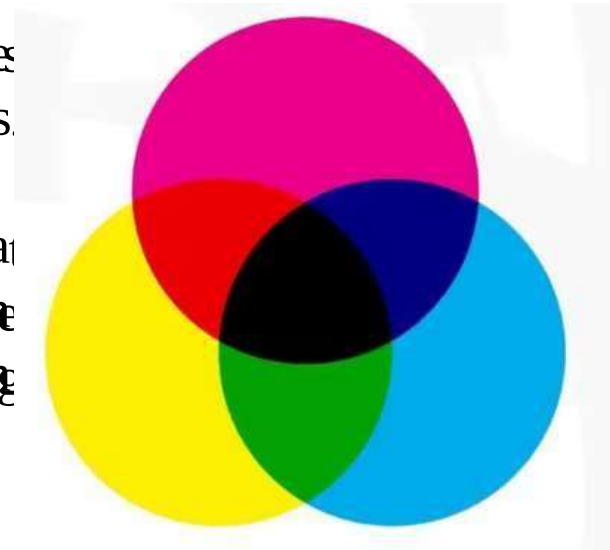
Printed images are a series of layered four-color dots (measured in dots per inch) that create different hues and gradations. Although CMYK is a standard color model, the exact range of colors represented can vary depending on the press and printing conditions.

Uses:

- Stationery (e.g., business cards).
- Advertising (e.g., posters, flyers, brochures).
- Product Packaging.

Tip—Use CMYK mode when preparing an image to be printed with process colors, as image conversion from RGB to CMYK creates a color separation. If you start with an RGB image, it's best to edit first in RGB and then convert to CMYK afterward.

Index: This mode produces **8-bit** image files with up to **256** colors. Like RGB, this color mode is exclusively for digital formats, on-screen. When you convert an image to index color, a color table gets built to store and index the image's colors





COLOUR MODES AND RESOLUTION OPTIONS

COLOR MODES

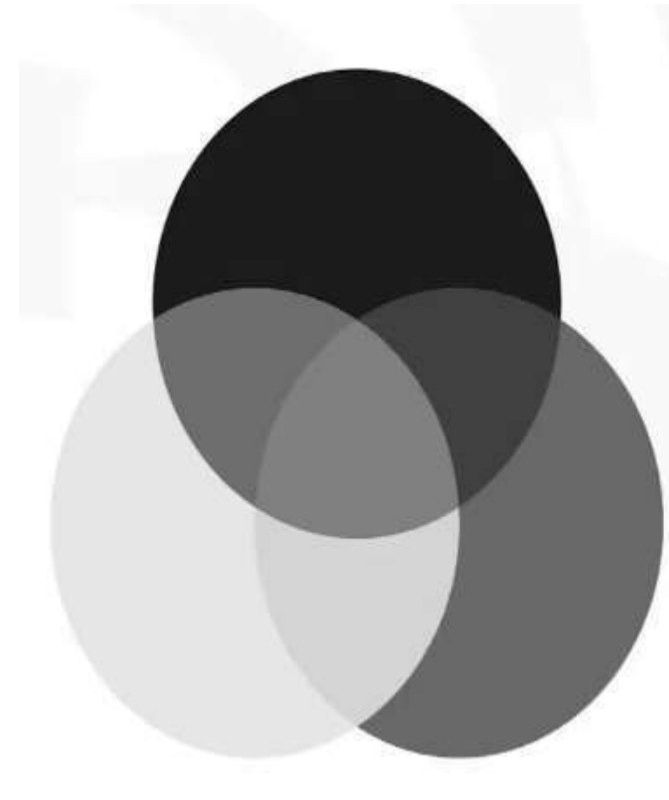
• Grayscale:

- Represents images **using shades of gray, from black to white.**
- This black-and-white-looking mode comprises various shades of grey within an image.
- We can use it in print and digital formats. In digital, each image pixel has a value ranging from 0 (black) to 255 (white). In print, the values range from 0% (white) to 100% (black), representing the amount of black ink used.

Uses:

- Print and digital formats to create a line drawing or hand-drawn sketch or make vintage effects.
- Digital formats to express a specific tone in your designs.

Print formats to lower costs and minimize ink usage.





COLOUR MODES AND RESOLUTION OPTIONS

COLOR MODES

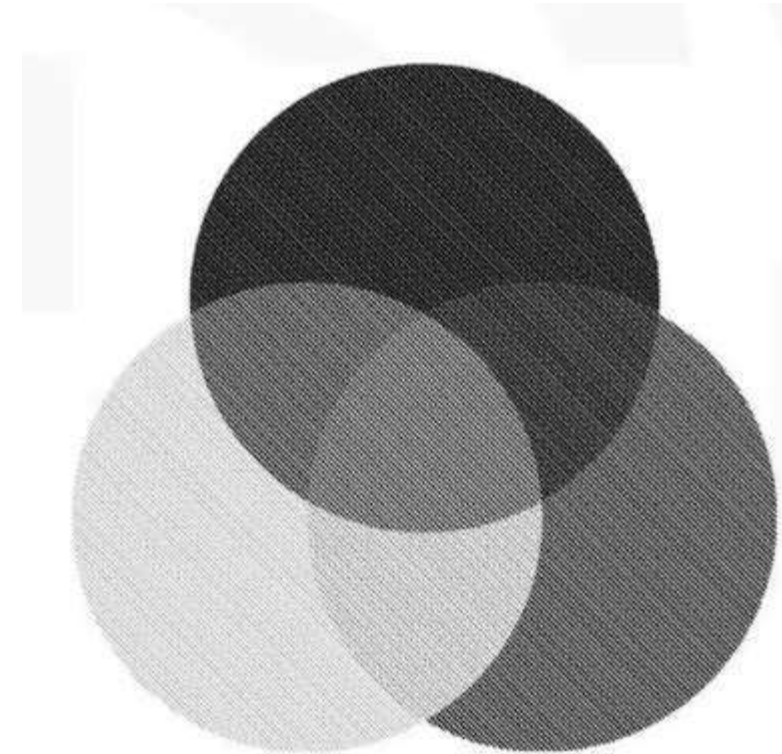
•Bitmap:

- Displays images with only two colors: black and white.
- **Bitmap** (. line art): This mode comprises black and white pixels, with no colors or shades of grey.

In digital formats, black and white values represent an image's pixels, while black ink dots and the white of the paper represent the overall image in print formats.

Uses:

Print and digital formats to create a line drawing or hand-drawn sketch or make vintage effects.





COLOUR MODES AND RESOLUTION OPTIONS

COLOR MODES

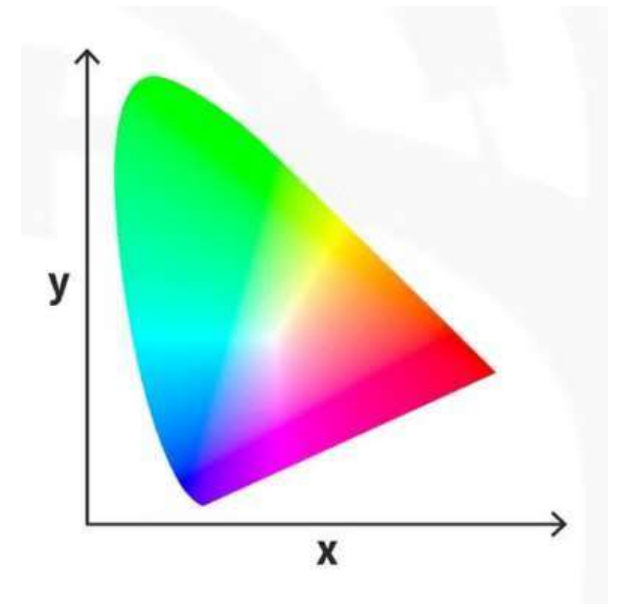
• Lab:

- A device-independent model based on how humans perceive color.
- Useful for color correction and ensuring consistency across different media

LAB color is ***device-independent***, making it easy to achieve *precisely* the same color across different types of media so your (e.g.) digital logo appears identically on a mug, banner, etc. However, LAB's large file sizes can delay load times.

Uses:

- Branded Products (e.g., T-shirts, banners).
- Color Reference.
- Photography.
- Improving images with a natural and vibrant look.





COLOUR MODES AND RESOLUTION OPTIONS

COLOR MODES

• Indexed Color:

- Uses a palette of up to 256 colors to reduce file size, making it suitable for web and multimedia.
- This mode produces **8-bit** image files with up to **256** colors. Like RGB, this color mode is exclusively for digital formats, on-screen.

Uses:

Websites. Digital [Presentations](#).

Mobile Applications.



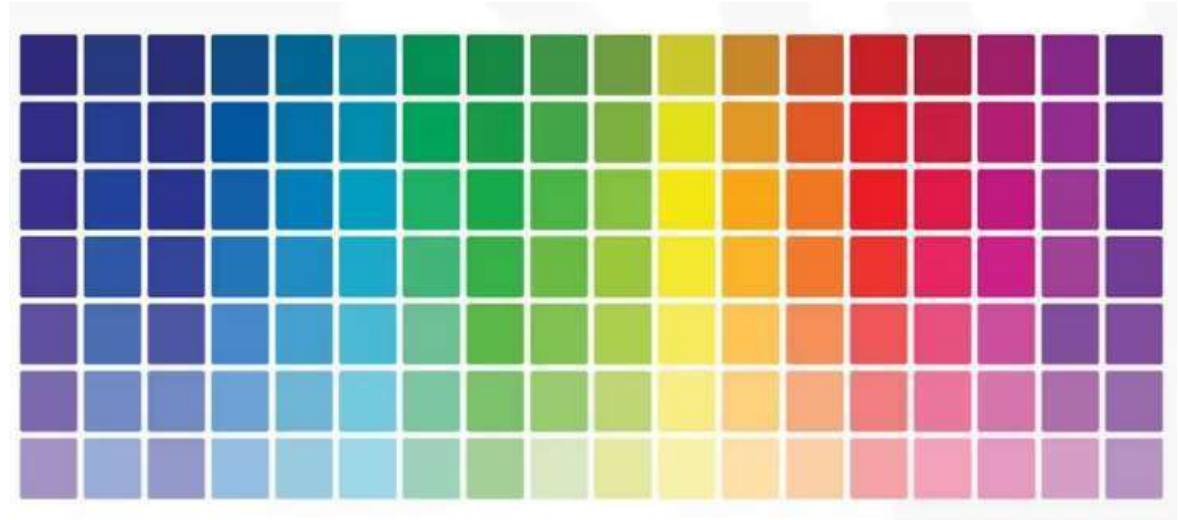
COLOUR MODES AND RESOLUTION OPTIONS

COLOR MODES

• Indexed Color:

Uses:

- Websites.
- Digital [Presentations](#).
- Mobile Applications.



While its [color palette](#) is limited, index color can reduce file size yet maintain the desired visual quality for digital presentations, websites and mobile applications. So, this mode is ideal for **image optimization**. For extensive editing, it's best to convert temporarily to RGB mode.



COLOUR MODES AND RESOLUTION OPTIONS

- The **display resolution** or display modes of a digital television, computer monitor, or other display device is the number of distinct pixels in each dimension that can be displayed.
- It can be an ambiguous term especially as the displayed resolution is controlled by different factors in cathode-ray tube (CRT) displays, flat-panel displays (including liquid-crystal displays) and projection displays using fixed picture-element (pixel) arrays.
- It is usually quoted as ***width × height***, with the units in pixels: for example, 1024×768 means the width is 1024 pixels and the height is 768 pixels. This example would normally be spoken as "ten twenty-four by seven sixty-eight" or "ten twenty-four by seven six eight".



COLOUR MODES AND RESOLUTION OPTIONS

COMMON DISPLAY RESOLUTION OPTIONS

- **HD (720p):** 1280×720 pixels. Suitable for smaller screens or basic displays.
- **Full HD (1080p):** 1920×1080 pixels. The standard for many desktop monitors, laptops, and TVs.
- **QHD (1440p):** 2560×1440 pixels. Offers a sharper image than Full HD and is often used for gaming.
- **4K UHD (2160p):** 3840×2160 pixels. Provides a much higher level of detail and is becoming more common in premium displays and TVs.
- **8K UHD:** 7680×4320 pixels. The highest resolution for current consumer displays.



FILTER EFFECTS

Definition:

A *filter effect* is a transformation applied to an image to change its appearance. Filters can enhance details, remove noise, or create artistic effects.

- Filter effects are **a way to apply visual transformations to an element or image before it is displayed, used in contexts like web design (CSS) and photography.**
- In CSS, they modify how an element renders, with effects like blur, brightness, and saturation, while in photography, filters can be digital presets or physical lenses that alter color, contrast, and other visual qualities for creative purposes.



FILTER EFFECTS

In web design (CSS)

•What they are:

Visual effects applied to elements using the filter or backdrop-filter CSS properties.

•Examples of effects:

- blur()**: Softens the image.
- brightness()**: Adjusts the brightness.
- contrast()**: Changes the contrast.
- grayscale()**: Converts the element to grayscale.
- hue-rotate()**: Shifts the colors in the image.
- invert()**: Inverts the colors.
- sepia()**: Gives a sepia-toned effect.

•Advantages:

They can be applied directly in the browser without needing external software or extra HTTP requests, and they are responsive and animatable.



FILTER EFFECTS

TYPES OF IMAGE FILTERS:

Filter Type	Description	Example Effect
Blur	Smooths transitions between colors or pixels, reducing detail.	Gaussian Blur, Box Blur
Sharpen	Enhances edges to make an image appear clearer.	Unsharp Mask
Edge Detection	Highlights boundaries and lines.	Sobel, Canny, Laplacian filters
Emboss	Creates a 3D shadow or raised effect.	Simulates texture
Grayscale	Removes color, leaving only brightness information.	Converts RGB → Gray
Sepia	Gives an old-fashioned brownish tone.	Vintage look
Invert	Reverses colors (light becomes dark, etc.).	Negative image
Saturation / Brightness / Contrast	Adjusts intensity of color, light, or tone.	Enhancing visibility or mood



FILTER EFFECTS

In photography

• **What they are:** Presets or physical filters that give an image a specific look or style.

• **Examples of effects:**

• **Soft/Mist:** Lowers contrast, softens skin tones, and makes light sources bloom or glow, creating a cinematic look.

• **Contrast/Sharpness:** Increases contrast and sharpness to make an image clearer.

• **Color filters:** Can be used to change the color temperature (warm or cool) or apply a specific color effect to the entire image.

• **How they work:** They can be applied digitally through software or by placing a physical filter in front of the camera lens.



FILTER EFFECTS

In photography

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Animation using Photoshop.

Photoshop supports **two main animation modes**:

➤ **Frame Animation (Timeline frames)**

Best for simple animations, GIFs, or stop-motion–style motion.
You control each frame individually (like a flipbook).

➤ **Video Timeline Animation**

Best for smooth, keyframe-based animations (like motion in After Effects).

You can animate layer positions, opacity, styles, and transformations over time.

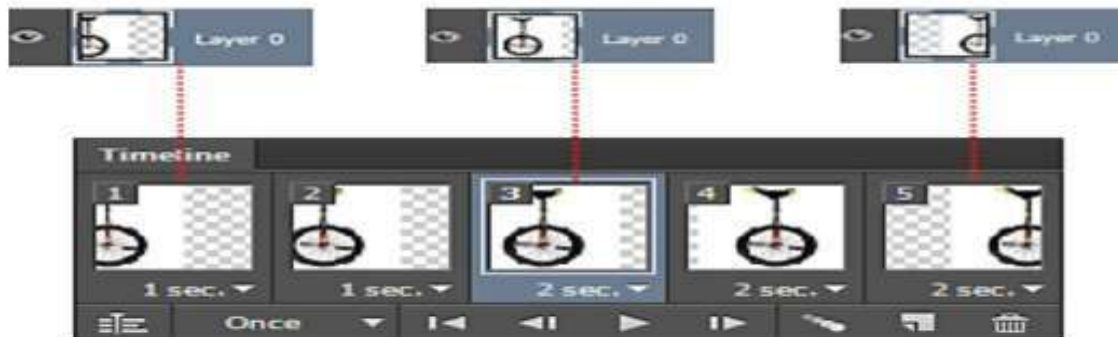


Illustration of an animation. The unicycle image is on its own layer; the position of the layer changes in each frame of the animation.



Animation using Photoshop.

Setting Up Your Animation

? Step 1: Create a New Document

Go to **File → New**

Choose your **canvas size** (e.g., 1080×1080 px for social media, 1920×1080 for HD video).

Set **resolution**: 72–150 ppi for web, 300 ppi for print.

Click **Create**.

? Step 2: Open the Timeline Panel

Go to **Window → Timeline**.

You'll see a small panel at the bottom.

Click **Create Frame Animation** or **Create Video Timeline** depending on the style you want.



Animation using Photoshop.

3. Creating Frame Animation

Basic Steps:

- **Add your layers** — Each object or element you want to animate should be on a separate layer.

In the **Timeline panel**, click **Create Frame Animation**.

- **Duplicate frames** by clicking the “+” (**New Frame**) icon.
- **Change the layer visibility or position** on each frame to create motion or changes.
- **Set frame delay time** below each frame (e.g., 0.1s, 0.5s).
- Click **Play** ► to preview.

Example:

Frame 1: Circle on the left.

Frame 2: Move the circle slightly right.

Frame 3: Move further right.

Result → smooth sliding motion.

Looping:



Animation using Photoshop.

4. Creating Video Timeline Animation (Keyframes)

Step 1: Switch to Video Timeline

In the Timeline panel menu, choose **Convert to Video Timeline**. You'll see each layer turn into a horizontal bar representing time.

Step 2: Add Keyframes

Expand the arrow beside the layer n

Position

Opacity

Style

Click the **stopwatch icon** [?] [?] to cre

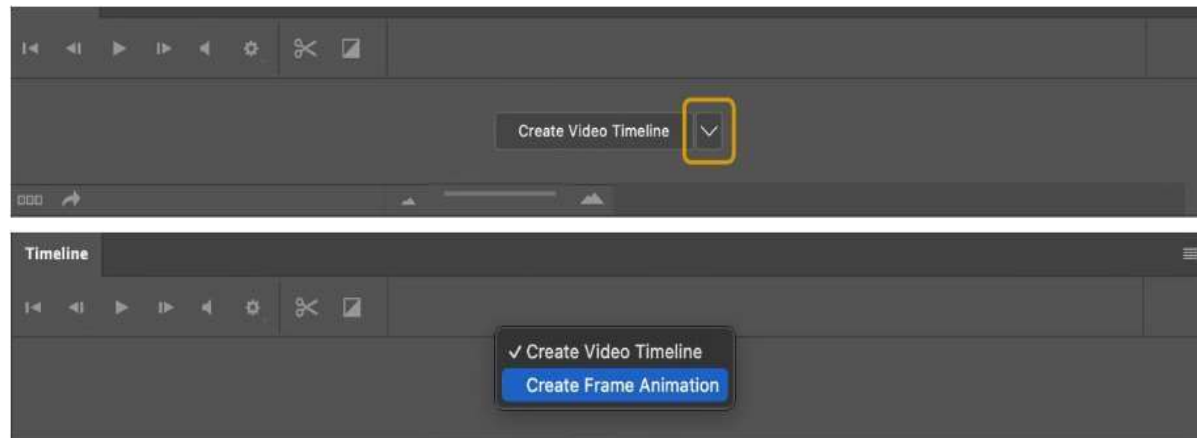
Move the playhead on the timeline
(move an object).

Photoshop automatically **interpolates** between the keyframes —
creating motion.

Step 3: Preview & Refine

Press **Spacebar** to
preview animation.

Animation by
frames.
Add
frames, or
frames for
seconds.





Animation using Photoshop.

4. Creating Video Timeline Animation (Keyframes)

Step 1: Switch to Video Timeline

In the Timeline panel menu, choose **Convert to Video Timeline**. You'll see each layer turn into a horizontal bar representing time.

Step 2: Add Keyframes

Expand the arrow beside the layer name → you'll see options like:

Position

Opacity

Style

Click the **stopwatch icon**   to create a keyframe.

Move the playhead on the timeline → adjust the property (e.g., move an object).

Photoshop automatically **interpolates** between the keyframes — creating motion.

